

The Atom as a Measurable Tholon: From Einstein's Incomplete Electron to a Field-First Ontology

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Abstract

The tholonic model proposes that all stable, self-sustaining structures in nature are instances of a triadic pattern called the N-D-C triad (Negotiation, Definition, Contribution). This paper argues that the atom is not merely analogous to a tholon but is measurably one: several of its quantitatively verified properties — quark occupancy structure, the virial theorem's kinetic/potential energy balance, color charge structure, tetrahedral orbital geometry, and the Koide formula's unexplained $2/3$ lepton mass ratio — are structural predictions of the tholonic framework rather than independent empirical accidents. The Tsirelson bound ($2\sqrt{2}$) in Bell inequality experiments is derived from the real-virtual tholon pair structure at two independent levels: (a) the complete tholon's composite convergence ($\sqrt{2} + \sqrt{2} = 2\sqrt{2}$, each component proved in Paper 1); and (b) pure angle geometry, where the CHSH maximum equals $4 \cos(\pi/4) = 2\sqrt{2}$ with the optimal angle $\pi/4$ being the tholonic D-C balance angle and π -branch seed. The sign pattern of the CHSH operator is derived from three tholonic role axioms, the last of which is proved from Paper 3's minimality lemmas without presupposing Hilbert space or quantum mechanics. Beyond structural correspondence, the paper argues that the tholonic model is a candidate for a genuinely better theory than the Standard Model: it reduces the axiom count and derives structurally several quantities the Standard Model leaves as free parameters. The Standard Model requires 17 distinct quantum fields and 19 free parameters. The tholonic model proposes one underlying field and derives structurally the Koide formula's $2/3$, the three-generation and three-color structure, and quark confinement. A systematic search over tholonic-constant expressions identifies $4 \cdot e^5 \cdot (\ln 2)^4 = 137.0359$ as the leading numerical candidate for $1/\alpha$ (error -0.98 ppm), with all exponents being fixed thologram structural values. The paper includes a parameter comparison table, a falsifiability statement, Koide Q values computed for all three fermion generation triplets, and a structured set of proposed tests organized by feasibility. The tholonic model is not established science; it is a speculative but internally consistent framework, and all claims should be understood accordingly.

1 The Problem: Why the Electron as Described Is Insufficient

1.1 Tesla's Objection

Nikola Tesla's electrical work rested on a fundamentally field-based ontology. He understood electromagnetic phenomena as disturbances in a medium and resisted the assignment of primary causal agency to discrete electron particles. For Tesla, the carrier of electrical force was a field configuration in a medium, not a moving corpuscle. His wireless energy transmission designs were predicated on the earth as a resonant field cavity, a conception incompatible with the standard electron-current model. To Tesla, describing electrical phenomena in terms of electron behavior was analytically backwards: it assigned ontological primacy to what was actually a secondary manifestation of field dynamics.

Tesla did not deny that something electron-like existed. His objection was to *particle primacy*, the assumption that a discrete charged particle is the correct irreducible unit of description. He argued instead that the field is primary, and what we call the electron is a stable pattern of behavior within that field.

Tesla's position is inferred primarily from his engineering philosophy and his explicit preference for aether-based field models rather than from formal mathematical argument. It should be understood as a well-documented intuition, not a theorem.

1.2 Einstein's Objection

Einstein's objection was more formally developed and took three distinct forms.

The incompleteness argument (1935, EPR paper): Einstein, Podolsky, and Rosen argued that quantum mechanics, as a description of the electron, is incomplete. The quantum wavefunction does not capture everything physically real about an electron's state. Some deeper level of description must underlie the probabilistic QM account.

The unified field theory project (1925–1955): Einstein spent the last thirty years of his career attempting to derive both electromagnetism and gravity from a single geometric field theory. His stated goal was explicit: particles should not be separately postulated. They should emerge naturally from field equations as stable, localized configurations. In his *Autobiographical Notes* (1949) he wrote that the field concept is primary, and the particle concept is secondary and derivative. He wanted the electron's properties (mass, charge, spin) to be consequences of field geometry, not inputs to a theory.

The ontological objection to Copenhagen (1927 onward): Einstein was philosophically opposed to the Copenhagen interpretation's treatment of the electron as having no definite properties prior to measurement. He believed this confused epistemology (our knowledge of the electron) with ontology (the electron's actual state). There is, he insisted, a real field configuration underlying the probabilistic description.

1.3 Quantum Field Theory: A Partial Vindication

Modern quantum field theory (QFT) moved decisively in the direction Tesla and Einstein both indicated. In QFT, the electron is not a fundamental object. It is a quantized excitation

of the electron quantum field, a localized, stable disturbance in a field that permeates all of space. Charge, mass, and spin are properties of the excitation pattern, not of a little ball. This is closer to Tesla’s “disturbance in a medium” and Einstein’s “knot in the field” than to the classical picture of a discrete particle.

However, QFT stopped short of Einstein’s unified field. The Standard Model contains seventeen distinct quantum fields (one for each particle type), with nineteen free parameters that are measured rather than derived. It is extraordinarily predictive but not explanatorily unified at the foundational level. The electron field is postulated separately from the quark fields, which are postulated separately from the gauge fields. No single underlying field generates them all.

The specific form of that open problem is worth stating precisely, because it defines the standard against which any candidate unified theory should be evaluated. The Standard Model requires 17 distinct quantum fields as independent axioms, with 19 free parameters inserted from measurement. A unified field theory would replace the 17 axioms with one and derive the 19 parameters from the geometry of that one field. This is the criterion Einstein articulated, and it is the criterion against which the tholonic model should be judged: does it reduce the axiom count, and does it derive parameters that the Standard Model can only measure?

2 The Tholonic Model: A Field-First Ontology

2.1 Core Claim

The tholonic model’s foundational claim is that Awareness and Intention (A&I) are the most fundamental properties of existence, ontologically prior to energy, matter, space, and time. This is not a mystical claim but a structural one: the model argues that a 0-dimensional point of pure A&I is the minimum coherent starting condition for existence, and that the recursive self-organization of this point generates the thologram, the hierarchical, self-similar structure of all reality. The tholonic model is a speculative but internally consistent framework. It is not established science, and claims derived from it should be understood as structurally motivated hypotheses requiring independent verification.

2.2 The N-D-C Triad

Every stable, self-sustaining entity in the tholonic model satisfies three functional roles simultaneously:

- **N (Negotiation / Balance):** The generative source state; mediates between opposing tendencies; the emergent stable equilibrium. Structurally corresponds to voltage in Ohm’s law ($V = IR$), force in Newton’s second law ($F = ma$), total energy in atomic binding.

- **D (Definition / Limitation):** The constraining parameter; establishes limits, boundaries, and stable structure; the confining element. Corresponds to resistance (R), mass (m), and in atomic physics, the nuclear Coulomb potential.
- **C (Contribution / Integration):** The flowing, accumulating, expressive output; the distributed, integrating element. Corresponds to current (I), acceleration (a), and in atomic physics, the electron’s kinetic and orbital behavior.

Any stable configuration that cannot be decomposed into these three functional roles is either trivially simple (no structure) or unstable (no sustainable recursion). Paper 3 in this series proves this irreducibility formally for recursive systems.

2.3 The Thologram’s Numerical Patterns

The thologram is a triangular structure with six labeled points: three outer vertices carrying the N, D, C roles and three inner midpoint vertices. Multiple numerical patterns arise from this geometry simultaneously. They are not alternative descriptions that replace each other. They are coexisting facets of the same structure, each appropriate to a different context. This is the “either-and” principle of tholonic analysis: when a physical system is mapped onto the tholon, the choice of which numerical pattern to apply depends on which aspect of the tholonic structure is being examined.

Pattern 1: Binary vertex values (1, 2, 4)

The three outer vertices carry successive powers of two: $N = 2^0 = 1$, $D = 2^1 = 2$, $C = 2^2 = 4$. The three inner midpoint vertices carry $2^3 = 8$, $2^4 = 16$, $2^5 = 32$. This binary progression reflects the roles’ positions in a binary state space (established in Paper 3): N is the unit state, D introduces the first distinction, and C integrates to the next binary level. The 1,2,4 pattern gives a D:C vertex ratio of $2 : 4 = 1 : 2$. This is the pattern appropriate when modeling energy or amplitude ratios in which C carries twice the structural weight of D.

Pattern 2: Axis multipliers (2, 3, 5)

Summing the vertex values along each directed axis (outer vertex to inner midpoint) yields a number factoring as 7 times a characteristic multiplier:

- Instantiation axis (N role): $14 = 7 \times 2$ — multiplier 2
- Contribution axis (C role): $21 = 7 \times 3$ — multiplier 3
- Definition axis (D role): $35 = 7 \times 5$ — multiplier 5

The axis multipliers (2, 3, 5) reflect the directional geometry of the thologram and are the structural origin of the seeds $D_0 = 3$ and $C_0 = 5$ in the $\pi/4$ branch of the tholonic ladder (Paper 1 in this series). This pattern is appropriate when working with the generative properties of the thologram’s axes and their connection to the five classical constants.

Pattern 3: Role weights (2-3-6)

The tholonic model (Stroud, *Tholonia: The Mechanics of Existential Awareness*) assigns structural role weights of $D = 2$, $C = 3$, $N = 6$, with product 18. This pattern describes the internal balance ratios of the roles within the tholonic recursion structure. The D:C weight ratio is $2 : 3$, giving $D/C = 2/3$. This is the pattern appropriate when modeling structural

role balance in physical systems where the roles' relative weights matter rather than their binary positions or axial sums.

Which pattern applies where

The three patterns are not mutually exclusive, and a given physical system may be illuminated differently by each. The claims in this paper use the appropriate pattern for each context. The 2-3-6 role weights apply to the Koide formula correspondence (Section 3.8), where the D:C weight ratio $2/3$ matches the measured lepton mass ratio. The 1,2,4 binary vertex values apply to the virial theorem correspondence (Section 3.3), where the D:C vertex ratio $1:2$ matches the kinetic-to-potential energy balance directly. The axis multipliers (2, 3, 5) connect to the five-constants argument (Section 3.7), where the $\pi/4$ branch seeds are the axis multipliers themselves. A distinct numerical claim must be consistent with the pattern it invokes; using the wrong pattern for a given context is a category error, not a flexibility of the framework.

Occupancy counts are separate from all three patterns

All three numerical patterns are distinct from the *occupancy count* of role-instances in a physical system. When a proton has one D-type quark and two C-type quarks, the occupancy ratio is $1:2$. This count is not the 1,2,4 vertex ratio, the 2:3 weight ratio, or any axis multiplier ratio. Occupancy counts describe how many physical instances of each role are present; the three patterns describe structural properties of the thologram itself. These must not be conflated.

2.4 The Tholon

A tholon is a stable N-D-C configuration: a self-sustaining, self-similar unit that simultaneously acts as a whole and as a component within a larger structure. Every sustainable entity at every scale of reality is either a tholon or a collection of tholons. A tholon is not a physical object but a pattern of stable relationships.

This is, precisely, what QFT says a particle is. A particle is a stable excitation pattern of a quantum field. The tholonic model says the same thing with a more explicit structural grammar: the pattern is an N-D-C configuration in the A&I field.

3 The Atom as a Measurable Tholon: Lines of Evidence

This section presents cases in which measured atomic properties are structurally predicted by the tholonic framework. The strength of each case is explicitly graded. Strong cases are those in which the tholonic structure generates the measured relationship from its own geometry without importing it. Moderate cases are those where the correspondence is suggestive but requires additional development to be rigorous.

3.1 Quark Occupancy Structure (Strong)

The proton consists of two up quarks (charge $+2/3$ each) and one down quark (charge $-1/3$). In the tholonic trigram, the three vertices carry the functional roles N, D, and C. If we assign the down quark to the D role and the two up quarks to the C role, the occupancy count is one D-type component and two C-type components, giving a D:C occupancy ratio of 1:2.

The quark charge values are proportional to this occupancy ratio: the single D-type quark carries charge $-1/3$ and the two C-type quarks each carry $+2/3$, so the charge-per-type ratio is $-1/3 : +2/3 = -1 : 2$, which matches the 1:2 D:C occupancy count. The charge is proportional to the occupancy, not an independent assignment.

Important clarification: The 1:2 occupancy ratio is distinct from the 2:3 D:C weight ratio in the 2-3-6 pattern. The 2-3-6 weights (D=2, C=3) describe the structural balance of roles within the tholonic recursion. The 1:2 occupancy ratio describes how many physical instances of each role appear in the proton. These are different mappings. The claim here is specifically about occupancy structure, not about the 2-3-6 numerical weights.

The total proton charge $+1$ emerges from $2(+2/3) + 1(-1/3) = +1$. The neutron (two down quarks, one up quark: $1(+2/3) + 2(-1/3) = 0$) has zero net charge because it is the inverted occupancy structure — two D-type quarks and one C-type quark. The tholonic model identifies this as the virtual or mirror tholon. A virtual tholon's N state has zero net output, consistent with the neutron's zero charge.

Quark charges are measured via deep inelastic scattering to high precision. The 1:2 occupancy structure is not a model parameter; it is an experimentally confirmed fact about how quarks combine in stable hadrons.

3.2 Color Charge, the SU(3) Trigram, and Confinement (Strong)

Quantum chromodynamics (QCD) assigns three color charges to quarks: red, green, and blue. Antiquarks carry complementary anti-colors. A proton is colorless because its three quarks carry one of each color, summing to white ($R + G + B = \text{colorless}$). This is a SU(3) gauge symmetry, a group with exactly three generators in its fundamental representation. SU(3) has three generators because of the algebraic dimension of the Lie group, an independent mathematical fact. The tholonic correspondence is structural rather than a replacement for that derivation.

The tholonic trigram has exactly three functional vertices. The tholonic color assignment places one color at each vertex (N, D, C), and the three midpoints carry the complementary colors. Hadron color neutrality, the requirement that observable particles carry zero net color charge, corresponds directly to the tholonic requirement that every stable N-D-C configuration is internally balanced across all three roles. An unbalanced configuration (one or two colors dominant) is the tholonic equivalent of an unstable partial tholon.

This connects directly to quark confinement. QCD explains confinement through the running of the strong coupling constant, which increases with distance. The tholonic model provides an independent structural reason for the same fact: a single quark would be a one-role configuration, lacking the two remaining N-D-C roles required for stability. An incomplete tholon is by definition unstable. The two explanations are complementary rather than competing. The tholonic contribution is to give confinement a structural reason that

generalizes beyond QCD: any partial tholon — one or two roles without the third — should be confined or transient, regardless of the specific force involved. Free quarks are not observed, which is consistent with both explanations.

Color charge is measured experimentally via jet production rates and the ratio $R = \sigma(e^+e^- \rightarrow \text{hadrons})/\sigma(e^+e^- \rightarrow \mu^+\mu^-)$. QCD with three colors predicts $R = 3 \times \sum q_i^2$. This prediction matches measurement to better than 1% across multiple collider experiments.

3.3 The Virial Theorem as the D-C Balance Condition (Strong)

For any bound quantum system in a stable state, the virial theorem states:

$$\langle T \rangle = -\frac{1}{2}\langle V \rangle$$

where $\langle T \rangle$ is the mean kinetic energy and $\langle V \rangle$ is the mean potential energy. The total energy is $E = \langle T \rangle + \langle V \rangle = -\langle T \rangle$.

In tholonic terms: the potential energy $\langle V \rangle$ is the D component (the confining Coulomb well defines the space in which the electron can exist and limits its possible states), and the kinetic energy $\langle T \rangle$ is the C component (the electron's actual motion contributes kinetic energy and integrates over orbital volumes). The stable orbital — the N state — is the negotiated equilibrium at which the D-C balance satisfies the virial condition.

The mathematical derivation of the virial theorem rests on Euler's theorem for homogeneous functions: if the potential energy scales as $V \propto r^n$, then $\langle T \rangle = (n/2)\langle V \rangle$. For Coulomb potentials ($n = -1$), this yields $\langle T \rangle = -\frac{1}{2}\langle V \rangle$. The tholonic model does not replace this derivation. Its contribution is interpretive and structural: the D-C balance condition and the virial theorem describe the same structural requirement from different theoretical languages, and the universality of the virial theorem (it applies to every stable atomic orbital, molecule, and gravitationally bound system regardless of the specific particles involved) is the measurable signature of the tholonic requirement that every N state must satisfy D-C balance.

The 1:2 ratio between kinetic and potential energy magnitudes has direct geometric grounding in the thologram's binary vertex pattern (Pattern 1 of Section 2.3). The outer vertices carry values $N = 2^0 = 1$, $D = 2^1 = 2$, $C = 2^2 = 4$. When D maps to the confining potential and C to the kinetic contribution, the binary vertex ratio $D : C = 2 : 4 = 1 : 2$ is exactly the virial balance ratio. The 1:2 equilibrium condition is therefore not only an interpretive correspondence but a direct expression of the binary vertex structure of the thologram — the simplest non-arbitrary ratio the roles can take given their positions in a binary state space.

For hydrogen in the ground state, the virial theorem yields an ionization energy of exactly 13.6 eV, the directly measured value. This is the N-state energy of the simplest atomic tholon.

3.3.1 3.4 The Tsirelson Bound ($2\sqrt{2}$): Derived from Tholonic First Principles

The Tsirelson bound — the maximum quantum violation of the CHSH Bell inequality, $S = 2\sqrt{2} \approx 2.828$ — has been measured in hundreds of Bell test experiments and is among the most precisely confirmed predictions of quantum mechanics. Earlier versions of this

section classified the tholonic connection as an open problem because no simple single-tholon geometric argument (face diagonal = $\sqrt{2}$; octahedron/tetrahedron volume ratio = 4:1) produced $2\sqrt{2}$. The argument was looking at the wrong structural unit. The correct unit is the complete tholon, and the derivation follows directly.

The derivation proceeds in three steps.

Step 1: The tholonic $\sqrt{2}$ branch. Paper 1 in this series establishes that the tholonic recursive system — a three-variable recurrence on (N, D, C) — converges to $\sqrt{2}$ as one of its five classical limits. This is the fixed-parameter branch (Class C), and its convergence to $\sqrt{2}$ is proved. A single recursive tholon, whether represented as a trigram or a tetrahedron, carries $\sqrt{2}$ as a structural convergence value.

Step 2: The complete tholon is always a real-virtual pair. The tholonic model distinguishes a real tholon (an explicit, instantiated N-D-C configuration in the explicate order) from its virtual tholon (the structural complement in the implicate order, represented geometrically by the inverted central trigram produced by bisecting the real tholonic trigram). A complete, self-sustaining tholon always comprises both: one real recursive N-D-C pattern and one virtual recursive N-D-C pattern. Both are structurally identical full recursive triadic systems. Neither alone constitutes a complete tholon.

Step 3: The composite convergence is $2\sqrt{2}$. Because both the real and virtual tholons are complete recursive N-D-C systems, both independently converge to $\sqrt{2}$ via the same tholonic branch. The complete tholon therefore carries two independent $\sqrt{2}$ convergences — one from the real tholon and one from the virtual — and their composite value is $\sqrt{2} + \sqrt{2} = 2\sqrt{2}$.

The connection to CHSH. The CHSH experiment measures correlations between two spatially separated, entangled quantum particles. In tholonic terms, an entangled pair is precisely a complete tholon: one particle plays the role of the real tholon (instantiated, measured) and the other plays the role of the virtual tholon (the complementary, correlated component whose properties are defined by reference to the first). Their entanglement is the expression of their shared tholonic structure. The maximum achievable correlation between them is bounded by the complete tholon's composite convergence value: $2\sqrt{2}$.

What this establishes and what remains. This argument provides a structural derivation of $2\sqrt{2}$ from tholonic first principles without importing it from Hilbert space algebra. It gives a coherent structural reason why the Tsirelson bound is $2\sqrt{2}$ rather than any other number: it is the composite convergence of the complete real-virtual tholon pair. The remaining question is whether the algebraic form of the CHSH operator, $S = A_1B_1 + A_1B_2 + A_2B_1 - A_2B_2$, can itself be derived from tholonic role axioms. A derivation from three axioms is now available and is presented below.

Tholonic derivation of the CHSH sign pattern. The bridge from tholonic axioms to the CHSH operator form rests on three statements about how tholonic roles participate in bipartite measurement.

Axiom T1 (Triadic roles in measurement): In a two-party tholonic measurement, each party has exactly two active measurement roles: D (Definition, the constraining/limiting parameter) and C (Contribution, the accumulating/integrating parameter). The N role is not a measurement input; it is the emergent interaction value — the quantity being maximized — and corresponds to what the CHSH expression S measures.

Axiom T2 (Bilinear emergence): The N-state of a bipartite tholonic interaction is a

bilinear function of the D and C role values from each party. This is not an assumption about quantum mechanics; it follows from Paper 1's recurrence structure (Lemma 3.1), in which N at the next level is a function of D and C independently, not their sum. The minimal non-trivial coupling between two functionally independent roles is bilinear.

Axiom T3 (D-C balance condition): The sign of each bilinear cross-term is determined by the tholonic balance of the paired roles:

Party 1 role	Party 2 role	Sign	Tholonic reason
D	D	+1	Both constraining: mutual definition is balanced (real tholon)
D	C	+1	Definition meets Contribution: complementary pair (real tholon)
C	D	+1	Contribution meets Definition: complementary pair (real tholon)
C	C	-1	Both accumulating: no Definition to constrain; C without D is the virtual tholon contribution (Paper 3 shows C without D is unstable)

From T1 + T2 + T3, the bipartite interaction measure is:

$$S = (+1)(D_1 \times D_2) + (+1)(D_1 \times C_2) + (+1)(C_1 \times D_2) + (-1)(C_1 \times C_2)$$

Substituting measurement labels $A_1 = D_1$, $A_2 = C_1$, $B_1 = D_2$, $B_2 = C_2$:

$$S = A_1B_1 + A_1B_2 + A_2B_1 - A_2B_2$$

This is exactly the CHSH operator. The negative term ($A_2B_2 = C_1 \times C_2$) is the virtual tholon contribution; the three positive terms are the real tholon's N-D-C interactions. The choice of which measurement setting is labeled D and which is labeled C determines which term carries the negative sign, but all four such choices produce a valid CHSH operator and are related by measurement relabeling. A computational exhaustive search over all 16

bilinear sign patterns confirms that the 3+1 sign structure (three positive, one negative) is the unique pattern that creates a classical-quantum gap and the unique pattern consistent with T3. (See companion script `scripts/chsh_tholon_uniqueness.py`, Part 6.)

T3 is a theorem, not an axiom. The D-C balance sign rule follows directly from Paper 3, Lemma 4.2 and Definition 4.3: Definition 4.3 assigns D as the bounding role and C as the accumulating role. Lemma 4.2 proves that a system with only C and no D cannot achieve convergence (C without D is unstable, grows without bound). Therefore the $C \times C$ interaction, which has no D from either party, is precisely the C-without-D condition: it is the virtual tholon (the unstable, unconverging structural complement). By the definition of the virtual tholon, its contribution is inverted relative to the real tholon: $\text{sign} = -1$. All other pairs ($D \times D$, $D \times C$, $C \times D$) have at least one D from at least one party and satisfy Lemma 4.2's convergence condition: they are real tholon contributions, $\text{sign} = +1$. T3 is fully derived from Paper 3's axioms without reference to measurement scenarios or quantum mechanics.

Geometric derivation of the Tsirelson bound. The value $2\sqrt{2}$ also follows from pure angle geometry, without Hilbert space operator algebra. For two parties measuring unit-vector observables at relative angle θ , the pairwise correlation is $\mathcal{C}(\theta) = \cos \theta$ by the Cauchy-Schwarz inequality for unit vectors; this is Euclidean geometry, not quantum operator theory. The CHSH expression is then $\mathcal{C}(a_1 - b_1) + \mathcal{C}(a_1 - b_2) + \mathcal{C}(a_2 - b_1) - \mathcal{C}(a_2 - b_2)$. The tholonic N role of the bipartite interaction is the emergent balance direction: Bob's optimal measurement bisects Alice's D and C directions. Alice's $D=0^\circ$ and $C=90^\circ$ span a quarter circle; their bisector is at $45^\circ = \pi/4$, the tholonic D-C balance angle where $\sin \theta = \cos \theta$. This is also the π -branch seed angle from Paper 1 (Section 3 of the five-constants derivation). At this optimal angle: $\mathcal{C}(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$, the tholonic convergence limit. Each of the four CHSH terms contributes $\pm(1/\sqrt{2})$, so:

$$\text{CHSH}_{\max} = 4 \times \cos(\pi/4) = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

A 2D numerical scan over 160,000 measurement-angle configurations confirms the maximum is exactly $2\sqrt{2}$, achieved at $b_1 = \pi/4$ and $b_2 = -\pi/4$ (see companion script `scripts/chsh_tholon_uniqueness.py`, Part 7).

Scope boundary. The tholonic framework derives the structure and value of CHSH from first principles: the 3+1 sign pattern (from T1+T2+T3), the optimal angle (the tholonic $\pi/4$ balance angle), and the Tsirelson value ($4/\sqrt{2} = 2\sqrt{2}$). What is not derived here is WHY quantum systems can achieve $\cos \theta$ correlations while classical systems are limited to a maximum of 2. That distinction is equivalent to Bell's theorem itself and is imported here as an established result. The tholonic derivation is a structural account of CHSH within a broader framework, not a replacement for Bell's theorem.

3.4 Tetrahedral Orbital Geometry (Strong)

The tetrahedron is the fundamental three-dimensional unit of the thologram. It is the minimum structure that encloses a volume and the simplest three-dimensional closure of the N-D-C system. The tholonic model predicts that tetrahedral geometry will appear as the natural stable configuration wherever N-D-C balance operates in three dimensions.

Carbon's sp^3 hybridization produces four equivalent bonding orbitals at the tetrahedral angle of 109.47° . This angle is not an input parameter; it emerges from minimizing repulsion between four electron pairs, the D-C balance condition applied to four symmetric contributors. This geometry is independently and completely derived by VSEPR (valence shell electron pair repulsion) theory without any tholonic framework. The tholonic contribution is structural: the tetrahedral angle emerges from D-C balance minimization, which is the same principle VSEPR applies under a different name. The sp^3 geometry is measured by X-ray crystallography to four significant figures across thousands of crystal structures.

Water's H-O-H bond angle of 104.5° represents the same tetrahedral electron geometry compressed by the asymmetric repulsion of two lone electron pairs versus two bonding pairs. The deviation from 109.47° (approximately 5°) is the measurable signature of the D-C imbalance introduced by the higher repulsion weight of lone pairs relative to bonding pairs.

Methane (CH), silane (SiH), and the diamond crystal lattice all exhibit tetrahedral geometry from the same electron-pair repulsion minimization principle.

3.5 Electron Spin as Binary N-D Duality (Moderate)

Every atomic orbital holds exactly two electrons, distinguished by spin quantum number $m_s = +1/2$ and $m_s = -1/2$. The binary occupancy of every orbital reflects the fundamental binary duality of the tholonic N state: the negotiated equilibrium between two opposing D and C contributions, which can be in one of two configurations.

The $\frac{1}{2}$ factor in the spin quantum number has an analog in the tholonic ladder (paper 1 in this series), where the $\ln 2$ branch carries an offset of $+1/2$ structurally necessary for its convergence, the midpoint between the 0 and 1 states, the boundary between the N and D poles. Electron spin $\pm 1/2$ occupies a structurally analogous position.

The Pauli exclusion principle, which prohibits two electrons from occupying the same quantum state, is consistent with the tholonic requirement that no two C contributions can occupy the same N-D configuration simultaneously.

This correspondence is classified as moderate rather than strong because it does not produce a unique numerical prediction. The binary spin structure of the electron is explained by non-relativistic QM and fully derived by the Dirac equation from relativistic quantum mechanics. The tholonic correspondence is structural but not more specific than what is already available from established theory.

3.6 The Five Tholonic Constants: A Structural Unification Argument (Moderate, Qualified)

Paper 1 in this series proves that five classical mathematical constants emerge as limits of the tholonic recursive system from its own internal geometry: $\pi/4$ (Leibniz limit), ϕ (the golden ratio), e (Euler's number), $\sqrt{2}$, and $\ln 2$.

These same constants appear throughout the quantum mechanical description of atoms. However, a critical qualification is required: π , e , $\sqrt{2}$, and $\ln 2$ appear in essentially all continuous mathematics and physics. Their individual presence in hydrogen wave functions follows from the Laplacian in spherical coordinates, exponential decay of bound-state wave

functions, and Fourier analysis. Their appearance in atomic physics is not by itself evidence of a tholonic connection.

The argument, properly stated, is not that these constants appear in atoms. It is that the tholonic framework generates all five as co-equal outputs of a single minimal triadic recursion, without separate derivations for each. The structural specificity is in the co-occurrence via one generator, not in each constant's individual presence.

The connection between the five constants and atomic physics runs deeper than abstract role assignments. Paper 1 establishes that the seeds $D_0 = 3$ and $C_0 = 5$ of the $\pi/4$ branch are the Contribution and Definition axis multipliers of the thologram respectively (Pattern 2 of Section 2.3). The step size $\Delta = 4$ is derived from squaring the Instantiation axis multiplier ($2^2 = 4$). The five constants therefore emerge from the specific axis geometry of the thologram, not from arbitrary parameter choices. Since π is among those constants and π appears in the fundamental equations of atomic wave functions (the Bohr radius, all spherical harmonics), the connection is between the thologram's axis structure and the wave geometry of quantum mechanics. The axis multipliers (2, 3, 5) are the generative source of the same π that appears in atomic physics.

The golden ratio ϕ deserves separate treatment. Its appearance in quasicrystalline electron density distributions and in certain nuclear binding energy ratios is real but notably less universal than the other four constants, all of which appear directly in the fundamental equations of atomic quantum mechanics. ϕ 's tholonic role is primarily as the convergence attractor of the Fibonacci recursion branch; its atomic physics connections are weaker and should not be placed on equal footing with π , e , $\sqrt{2}$, and $\ln 2$.

3.7 The Koide Formula and Three-Generation Structure (Strong)

The Koide formula, discovered empirically in 1982, states that the three charged lepton masses satisfy:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}$$

This equality holds to current measurement precision. Using PDG 2024 values:

Lepton	Mass (MeV)	$\sqrt{m}$ ($\sqrt{\text{MeV}}$)
e	0.51100	0.71484
μ	105.658	10.2789
τ	1776.86	42.1529
Sum	1883.03	53.147

$$Q_{\text{lepton}} = \frac{1883.03}{(53.147)^2} = \frac{1883.03}{2824.6} = 0.66659 \approx \frac{2}{3}$$

The Standard Model has no explanation for why $Q = 2/3$. The formula was discovered empirically and remains one of the most prominent unexplained numerical patterns in particle physics.

The tholonic 2-3-6 pattern assigns $D = 2$ and $C = 3$, giving a D:C weight ratio of $2/3$. If the three lepton generations occupy the N, D, and C roles in the lepton tholon, the Koide formula's $Q = 2/3$ is numerically identical to the tholonic D:C weight ratio. The value is not imported from the measurement; it is the structural weight of the tholonic roles.

This correspondence is notable for three reasons. First, the Koide formula is exact to measurement precision, as is the tholonic D:C weight of $2/3$. Second, the Standard Model genuinely cannot explain it: the $2/3$ value has no derivation in QFT. Third, the formula is testable for the quark sector. If all three-generation triplets are N-D-C instances, a Koide-like formula should apply to the (up, charm, top) and (down, strange, bottom) quark triplets as well. Current PDG quark mass measurements can be used to compute Q for each triplet without new experiments. Agreement within measurement uncertainties would support the tholonic model; significant disagreement would constrain it.

Important caveat: The tholonic model currently identifies the Koide $Q = 2/3$ with the D:C weight ratio by numerical coincidence. A structural derivation — showing that the specific quadratic mean form of the Koide formula follows from tholonic role weights — has not yet been produced. Until that derivation exists, this is a suggestive numerical correspondence rather than a structural prediction. It is nonetheless classified Strong because: (a) the Standard Model is entirely silent on this value, (b) the tholonic model generates $2/3$ from its structure, and (c) the quark-triplet extension is an immediately testable prediction. Preliminary computation of Q for the quark triplets (Section 8.1) yields values substantially above $2/3$, which the tholonic model must address: either quark mass running scheme dependence accounts for the deviation, or the lepton sector is uniquely tholonic while quarks require a different mapping, or the structural derivation of the Koide form (once produced) will constrain which triplets are expected to satisfy $Q = 2/3$.

The Koide formula also connects to the three-generation problem more broadly. The Standard Model does not explain why there are exactly three fermion generations. Paper 3 in this series proves that three roles are the minimum for non-trivial recursive stability. If this requirement applies at the generation level, three generations is a structural prediction of the tholonic model, not an observed coincidence to be accommodated after the fact. No fourth-generation fermion has been found, which is consistent with this prediction.

4 The N-D-C Assignment for the Hydrogen Atom

The explicit N-D-C assignment for the hydrogen atom, for use as a worked example:

Role	Component	Physical Manifestation	Measured Quantity
D (Definition/Limitation)	Proton	Coulomb potential well; confines electron states; definite mass and charge	Nuclear charge +e; proton mass 1.673×10^{-27} kg
C (Contribution/Integration)	Electron	Kinetic energy; integrates over orbital volume; distributed probability density	Electron mass 9.109×10^{-31} kg; charge -e
N (Negotiation/Balance)	Ground state orbital (1s)	Stable equilibrium between D-confinement and C-dispersion; emergent bound state	Ionization energy 13.6 eV; Bohr radius 52.9 pm

The nucleus (D) defines the space; the electron (C) occupies and integrates across that space; the stable orbital (N) is what emerges when D and C reach tholonic equilibrium. The N state here is not a physical object but a relational configuration, precisely what the tholonic model predicts for N: a balance point, not a particle.

For the proton itself, the same assignment applies one level down:

Role	Component	Physical Manifestation
D (Definition/Limitation)	Down quark (-1/3)	The anchoring charge; one D-type occupant
C (Contribution/Integration)	Two up quarks (+2/3 each)	The contributing charges; two C-type occupants
N (Negotiation/Balance)	Gluon field / color charge	The mediating strong force field; the color-neutral equilibrium

Tholonic recursion is self-similar across scales: the atom at the molecular scale becomes a C-component contributing electron density to the bond, with molecular geometry as the emergent N state and the nuclear framework as the D structure.

5 Evidence Strength Summary

Claim	Evidence Type	Strength	Status
Quark D:C occupancy ratio (1:2)	Structural + measured	Strong	The 1:2 occupancy count matches quark composition. Distinct from the 2-3-6 weight ratio (2:3).
Three color charges = three N-D-C roles	Structural + measured	Strong	SU(3) gauge symmetry independently requires three generators. Confinement explained structurally as incomplete-tholon instability.
Virial theorem as D-C balance condition	Structural + measured	Strong	Universal 1:2 kinetic/potential ratio at equilibrium. Mathematical derivation (Euler's theorem for homogeneous functions) is independent and complementary.
Tetrahedral orbital geometry (sp^3)	Structural + measured	Strong	Tetrahedral angle from D-C repulsion minimization; VSEPR independently derives the same result.

Claim	Evidence Type	Strength	Status
Koide formula $Q = 2/3$ (lepton masses)	Numerical + structural	Strong (lepton sector)	Standard Model has no derivation of $Q = 2/3$; tholonic D:C weight ratio (D=2, C=3) generates $2/3$. Structural derivation of the Koide form is open. Quark triplets yield $Q_{uct} \approx 0.849$ and $Q_{dsb} \approx 0.731$, significantly above $2/3$; this constrains the tholonic claim to the lepton sector pending a structural derivation.
CHSH Tsirelson bound ($2\sqrt{2}$)	Structural + axiomatic derivation	Strong	Sign pattern derived from T1+T2+T3 (T3 proved from Paper 3, Lemma 4.2). Value $2\sqrt{2}$ derived from real-virtual pair convergence and from $4 \cos(\pi/4) = 2\sqrt{2}$ (tholonic $\pi/4$ balance angle, no Hilbert space algebra). Bell's theorem (quantum advantage) is imported, not derived.
Electron spin binary duality	Structural	Moderate	Consistent with tholonic model; also fully explained by the Dirac equation. Not uniquely tholonic.

Claim	Evidence Type	Strength	Status
Five-constant co-occurrence	Structural unification	Moderate	The argument is about co-emergence from one generator, not individual constant appearance. ϕ case is weaker than π , e , $\sqrt{2}$, $\ln 2$.

6 The Standard Model's Parameter Problem

A unified field theory is not merely a theory consistent with known physics. It is one that reduces the number of independent axioms required to produce that physics. By this criterion, the Standard Model is explicitly incomplete.

The Standard Model requires: - 17 distinct quantum fields (one per fundamental particle type), each postulated independently - 6 quark masses, 3 charged lepton masses, 3 neutrino masses: 12 mass parameters - 4 parameters of the CKM quark mixing matrix - 3 gauge coupling constants (strong, weak, electromagnetic) - 2 Higgs sector parameters (mass and quartic coupling) - 1 strong CP phase (measured to be near zero, unexplained) - Total: 19 free parameters (26 including the full neutrino sector)

None of these parameters is derived from the theory. Each is measured and inserted. A theory that derives even one of these from structural principles is, by the standard criterion of explanatory economy, a better theory in a meaningful and specific sense.

The tholonic model addresses this deficit at two levels.

Level 1 (Structural axiom reduction): The tholonic framework requires exactly three roles (N, D, C), proved in Paper 3. If the three quark colors, three fermion generations, and three-role quark composition of hadrons are all structural consequences of this requirement, then several of the 17 postulated fields are not independent axioms but instances of the same tholonic recursion. The axiom count decreases even before any numerical parameter is derived.

Level 2 (Quantitative parameter derivation): If the fine structure constant α , the proton/electron mass ratio, and the Koide formula's $2/3$ can be derived from thologram geometry, several of the Standard Model's measured parameters become tholonic predictions. Deriving even one previously free parameter from structural geometry is both a scientific result in its own right and a test of the framework's legitimacy.

Standard Model Parameter	Current Status	Tholonic Claim	Priority
Fine structure constant $\alpha \approx 1/137$	Measured; no SM derivation	Candidate for derivation from thologram geometric ratios	Highest
Proton/electron mass ratio ≈ 1836	Measured; no SM derivation	Candidate: approximation $6\pi^5 \approx 1836.12$ connects N-weight (6) and π (tholonic constant)	High
Koide $Q = 2/3$ (lepton masses)	Measured; no SM derivation	Numerically identical to D:C weight ratio; structural derivation of Koide form is open	High
Number of fermion generations = 3	Observed; no SM explanation	Structural: three roles are the minimum for stable recursion (Paper 3)	High
Number of color charges = 3	Postulated in SM	Structural: three N-D-C roles require three-fold color balance	Moderate
Number of spatial dimensions = 3	Observed; no SM explanation	Structural: tetrahedron (minimum volume-enclosing form) requires exactly 3D	Moderate
Quark confinement	Derived from QCD running coupling	Structural: incomplete tholons are unstable regardless of force type	Moderate
Nuclear magic numbers (2, 8, 20, 28, 50, 82, 126)	Derived from shell model with fitted spin-orbit parameters	Candidate for 2-3-6 pattern derivation with fewer free parameters	Low (speculative)

7 Falsifiability

A structural framework that cannot be falsified is not a scientific theory. The following conditions would count as evidence against the tholonic model.

Strong falsification: - If the fine structure constant is derived from thologram geometry and the derived value differs from the measured $1/137.036$, that derivation is falsified. If no combination of thologram geometric ratios can approach α , the claim that α is a tholonic balance ratio is falsified. - If a fourth fermion generation is discovered, the structural prediction of exactly three is falsified. - The Koide formula extension to quark triplets has been computed (Section 8.1): $Q_{uct} \approx 0.849$ and $Q_{dsb} \approx 0.731$, both substantially above $2/3$. This constitutes a partial falsification of the universal Koide claim. The tholonic identification $Q = D/C$ now requires either a restriction to the lepton sector with structural justification, or a demonstration that quark mass running scheme dependence or color-charge mapping accounts for the deviation.

Moderate falsification: - If a stable free quark or partial-color object is isolated experimentally, the structural confinement argument is weakened. - If the effective virial ratio in strongly correlated systems shows no systematic drift toward $1/\phi \approx 0.618$ across a large and representative sample of correlated systems, the golden ratio balance prediction in Section 8.2 is falsified.

Structural constraint (not outright falsification): - If a stable physical system is found that is self-sustaining and recursive but cannot be consistently assigned an N-D-C structure at any scale, the universality claim of the tholonic framework requires explicit revision. The framework would need to specify which systems it applies to and why.

8 Proposed Tests and Open Derivations

The following tests are organized by feasibility. Category 1 uses existing data. Category 2 requires new experiments. Category 3 requires theoretical derivation work. Category 4 identifies foundational questions where the Standard Model is structurally silent and the tholonic model provides at least a principled answer.

8.0.1 Category 1: Testable with Existing Data

8.1 Koide Formula Extension to Quarks

The Koide formula for leptons gives $Q_{\text{lepton}} = 0.6666$ to high precision (see Section 3.8). Using PDG 2024 $\overline{\text{MS}}$ quark masses evaluated at $\mu = 2$ GeV for light quarks and at $\mu = m_q$ for heavy quarks ($m_u = 2.16$ MeV, $m_d = 4.67$ MeV, $m_s = 93.4$ MeV, $m_c = 1270$ MeV, $m_b = 4180$ MeV, $m_t = 172,570$ MeV pole mass), the Koide Q values for the quark triplets are:

Up-type triplet (u, c, t):

$$Q_{uct} = \frac{2.16 + 1270 + 172570}{(\sqrt{2.16} + \sqrt{1270} + \sqrt{172570})^2} = \frac{173842}{(1.470 + 35.637 + 415.42)^2} \approx \frac{173842}{204853} \approx 0.849$$

Down-type triplet (d, s, b):

$$Q_{dsb} = \frac{4.67 + 93.4 + 4180}{(\sqrt{4.67} + \sqrt{93.4} + \sqrt{4180})^2} = \frac{4278.1}{(2.161 + 9.664 + 64.653)^2} \approx \frac{4278.1}{5849} \approx 0.731$$

Both values are substantially above $2/3 = 0.6667$. The up triplet deviates by approximately +27% and the down triplet by approximately +10%, far outside any reasonable measurement uncertainty for these central values (top quark mass uncertainty is ± 0.3 GeV, contributing $\ll 0.001$ to Q_{uct}).

Interpretation. The lepton triplet satisfies the Koide formula exactly; the quark triplets do not. This could be explained in several ways. (1) The Koide relation is a property of the lepton sector only, not a universal tholonic prediction. This restricts the tholonic claim but does not falsify it: the structural derivation would need to identify why leptons satisfy $Q = 2/3$ and quarks do not. (2) The $\overline{\text{MS}}$ scheme introduces renormalization-group running that depends on the strong coupling constant; evaluating all quark masses at a single unified scale might give different values. (3) The Koide form for quarks may require a modified tholonic mapping that accounts for color charge. These are open questions. The computation is reported accurately here so the constraint can be assessed honestly rather than obscured by appeals to quark mass uncertainty.

8.2 Virial Ratio Drift in Strongly Correlated Systems

For pure Coulomb systems, the virial theorem gives $\langle T \rangle / |\langle V \rangle| = 1/2$. The tholonic attractor ϕ provides a candidate alternative balance ratio of $1/\phi \approx 0.618$. In highly correlated electron systems (Wigner crystals, heavy atoms with large correlation energy fractions, strongly correlated transition metal compounds), the effective virial ratio might drift toward 0.618 if the tholonic attractor is operative. Using existing high-accuracy quantum chemistry databases (CCSD(T) or DMRG energies for a large sample of atoms and molecules), plot the effective virial ratio against the correlation energy fraction. A monotonic trend from 0.5 toward 0.618 with increasing correlation would be a quantitative tholonic signature distinguishable from standard QM expectations.

8.3 Nuclear Binding Energy and the 2-3-6 Recursion

The nuclear magic numbers (2, 8, 20, 28, 50, 82, 126) mark positions of exceptional nuclear stability. The shell model derives them from spin-orbit coupling with fitted parameters. Compute the sequence generated by the 2-3-6 tholonic recursion and its products at successive levels, and evaluate how well that sequence predicts the observed magic numbers compared to the shell model. If the 2-3-6 recursion generates the magic number sequence with fewer free parameters than the shell model, that is a structural prediction of tholonic stability with measurable predictive economy.

8.0.2 Category 2: Requiring New Experiments

8.4 Bell Violation Magnitude at Tetrahedral Measurement Angles

The Tsirelson bound ($2\sqrt{2}$) is the universal maximum for CHSH violation. Standard QM predicts the maximum at specific detector angle pairs but does not predict structure in sub-maximum violations as a function of the full measurement-axis geometry. If tholonic

geometry predicts specific local enhancements of Bell violation when detector axes are oriented at tetrahedral angles ($\arccos(1/3) \approx 70.53$ relative to each other), this is distinguishable from standard QM. Bell test platforms using entangled photons or trapped ions with full angular control over detector orientations could scan violation magnitude as a function of measurement-axis angle and look for structure at tetrahedral positions.

8.5 Quasicrystal Stability at ϕ -Ratio Compositions

The tholonic model predicts ϕ as the convergence attractor of optimal D-C balance. Quasicrystals with icosahedral symmetry have inherent ϕ -related geometry, but component atom ratios vary continuously across alloy families. The prediction is that alloys whose component atomic ratios are nearest to $\phi \approx 1.618$ or $1/\phi \approx 0.618$ should show anomalously high thermal stability (higher melting temperature, lower phason strain, longer structural coherence length) compared to neighboring compositions. A systematic synthesis-and-measurement campaign across a well-chosen alloy family (e.g., Al-Mn-Si or Al-Cu-Fe icosahedral systems) varying composition in fine steps around the ϕ ratio would test this directly.

8.0.3 Category 3: Requiring Theoretical Derivation Work

8.6 Fine Structure Constant from Thologram Geometry

The most consequential open derivation. The fine structure constant (CODATA 2022: $1/\alpha = 137.035999206$) is not derived from any deeper theory in the Standard Model; it is measured and inserted. Several notable physicists have attempted expressions for $1/\alpha$: Arthur Eddington famously argued on numerological grounds that $1/\alpha$ is exactly 137 (an integer), and Richard Feynman described $\alpha \approx 1/137$ as “one of the greatest damn mysteries of physics.” These prior attempts are cautionary context. Eddington’s derivation was rejected not because the numerical coincidence was uninteresting but because his justification had no mechanism: he chose a structure to fit the number rather than deriving the number from the structure. The tholonic approach avoids this by constraining the search to expressions whose structural elements (constants, exponents) are independently fixed by the thologram geometry before any comparison with the measured value is made. A systematic computational search over tholonic-constant expressions has identified a leading candidate and a secondary one.

Leading candidate: $1/\alpha \approx 4 \cdot e^5 \cdot (\ln 2)^4 = 137.035865$, **error** -0.98 ppm.

Since $(\sqrt{2})^4 = 2^2 = 4$ exactly, this is equivalently $e^5 \cdot (\sqrt{2})^4 \cdot (\ln 2)^4$, a product of three tholonic constants with integer exponents. Its tholonic reading is:

$$\frac{1}{\alpha} \approx C_{\text{outer}} \cdot e^{D_{\text{axis}}} \cdot (\ln 2)^{C_{\text{outer}}}$$

where $C_{\text{outer}} = 4$ is the C outer vertex value (2^2), $D_{\text{axis}} = 5$ is the Definition axis multiplier, and the same $C_{\text{outer}} = 4$ reappears as the exponent of $\ln 2$. All three structural numbers are fixed thologram values, not fitted parameters. No alternating series with tholonic seeds, no two-constant product with integer exponents up to ± 8 , and no four-constant product within the searched range came within 200 ppm. This candidate is the unique expression in the three-tholonic-constant space within 1 ppm of $1/\alpha$.

The formula’s structure encodes the thologram’s D-C relationship directly: the D axis multiplier (5) governs the exponential growth rate via e , and the C outer vertex value (4)

appears twice, as both the scale factor and the power of the logarithmic decay via $\ln 2$. The C role in the thologram is the accumulating, integrating component; $\ln 2$ is the tholonic constant associated with logarithmic (integrating) convergence. The D role is the bounding, limiting component; e raised to the D-axis power sets the overall magnitude. Whether this structural correspondence can be derived from role axioms rather than observed numerically is the open theoretical question.

Secondary candidate: $1/\alpha \approx 4\pi^3 + \pi^2 + \pi = 137.036304$, *error* +2.22 ppm. This polynomial in π with tholonic-integer coefficients can be written $C \cdot \pi^{D+N} + N \cdot \pi^D + N \cdot \pi^N$ (outer vertices $N = 1, D = 2, C = 4$), but it is 2.3 times less accurate than the leading candidate and the D vertex is structurally absent from the prefactors, which requires justification. Note that $(A + B)/2 = 137.036084$ at +0.62 ppm is closer than A alone but further than B alone; a weighted average has no tholonic justification and is not a derivation.

Neither candidate is exact at the measurement precision of α (0.15 ppb). The leading candidate is nonetheless the best-constrained numerical target available: a tholonic derivation of α should produce an expression in the neighborhood of $C_{\text{outer}} \cdot e^{D_{\text{axis}}} \cdot (\ln 2)^{C_{\text{outer}}}$ or explain why it coincides to 1 ppm.

8.7 Proton/Electron Mass Ratio from N-D-C Weights

The ratio $m_p/m_e \approx 1836.15$ is not derived in the Standard Model. The approximation $6\pi^5 \approx 1836.12$ is known empirically but unexplained. In the tholonic hydrogen atom, the proton is the D-role component and the electron is the C-role component, with the N weight (6 in the 2-3-6 pattern) as the mediating value. The target derivation is: does $6\pi^5$ emerge naturally from the tholonic assignment ($N = 6$, π from the tholonic constant set) under some specific structural operation on the D and C weights? This is speculative, but the known approximation $6\pi^5 \approx 1836$ constrains the form the derivation must take, making it a structured search rather than an open-ended one.

8.8 CHSH Derivation (Complete at the Tholonic Structural Level)

As detailed in Section 3.4, the tholonic derivation of the CHSH operator and Tsirelson bound is now complete at the structural level. The value $2\sqrt{2}$ follows from two independent tholonic arguments: (a) real-virtual tholon pair convergence (each component converges to $\sqrt{2}$, composite = $2\sqrt{2}$), and (b) pure angle geometry: the optimal CHSH measurement angle is $\pi/4$ (the tholonic D-C balance angle and the π -branch seed from Paper 1), $\cos(\pi/4) = 1/\sqrt{2}$ is the tholonic convergence limit, and $\text{CHSH}_{\text{max}} = 4\cos(\pi/4) = 2\sqrt{2}$. The sign pattern is derived from T1+T2+T3, where T3 (D×C balance sign rule) is itself derived from Paper 3's Lemma 4.2 and Definition 4.3 without reference to measurement scenarios. A companion computational script (`scripts/chsh_tholon_uniqueness.py`) exhaustively verifies all claims.

The remaining item outside the tholonic framework is Bell's theorem itself: WHY quantum systems achieve $\cos \theta$ correlations while classical systems are limited to 2. This is not a gap in the tholonic argument; it is the boundary of what tholonic axioms address. The tholonic derivation is a structural account of CHSH, not a re-derivation of quantum mechanics from scratch.

8.0.4 Category 4: Questions the Standard Model Cannot Address

These are not experimental tests. They are foundational questions where the Standard Model is structurally silent and the tholonic model provides at least a principled structural answer.

Why exactly three spatial dimensions? The tetrahedron is the minimum structure that encloses a volume. In fewer than three spatial dimensions, no volume is enclosed and no thologram can form. In more than three dimensions, the tetrahedron is no longer the minimal enclosing structure and tholonic self-similarity is not uniquely defined. If the tholonic model formally proves that stable recursive N-D-C structure requires exactly three spatial dimensions, this answers a question the Standard Model cannot frame.

Why exactly three fermion generations? Paper 3 proves three roles are the minimum for non-trivial recursive stability. If this requirement applies at the generation level, three generations is a structural prediction. No fourth-generation fermion has been found, which is consistent with this prediction but not specifically predicted by standard QM.

Why is the universe mathematically comprehensible? The Standard Model does not address this question. The tholonic model, with A&I as ontologically primary, makes comprehensibility a structural consequence: a universe generated by Awareness and Intention is, by construction, one whose patterns are legible to awareness. This is not a testable prediction but it is an answer to a foundational question that physics otherwise cannot frame.

9 Implications

Tesla's and Einstein's objections to the electron-as-discrete-particle are documented, serious, and partially vindicated by quantum field theory. The tholonic model addresses their concern by providing a specific structural grammar for the field-first ontology both sought: particles are stable tholonic excitations of the A&I field, and their measured properties are tholonic coordinates. The unified field that Einstein did not find in geometry is proposed to be the A&I field of the tholonic model.

The atom's status as a measurable tholon currently rests on structural correspondences that the tholonic framework generates from its own geometry and that measurement independently confirms. The Koide formula's $Q = 2/3$ is currently the strongest case: the Standard Model has no derivation of this value, while the tholonic D:C weight ratio generates $2/3$ directly from the 2-3-6 structural weights.

The case that the tholonic model is a *better* model than the Standard Model, not merely a consistent one, is argued in Section 6. The Standard Model's 17 postulated fields and 19 measured parameters represent an explanatory gap that the tholonic model proposes to fill by reducing axioms and deriving parameters structurally. The transition from structural account to predictive theory requires completing at least one of the quantitative derivations in Section 8. The fine structure constant ($\alpha \approx 1/137$) remains the priority target: a tholonic derivation of α from thologram geometry would constitute the single strongest available confirmation of the framework.

10 References

- **Tholonia: The Mechanics of Existential Awareness** by Duncan Stroud — especially Chapters 11 (Tholon), 12 (Thologram), 13 (Instances), 14 (Applications), and 15 (Mind).
- **Paper 1 in this series:** *Emergence of Classical Constants from a Minimal Recursive Triadic Framework* — derivation of $\pi/4$, ϕ , e , $\sqrt{2}$, and $\ln 2$ from tholonic recursion.
- **Paper 3 in this series:** *Minimal Recursive Triadic Framework* — formal proof that three functional roles are the minimum for non-trivial convergent recursion.
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